

Cognitive Science — Module 05: Action — Study Questions

1. How does Active Inference explain action differently from traditional reward-based accounts?
2. What does it mean to say "action makes the world match the model"?
3. What is Expected Free Energy (EFE)? Explain it without using equations.
4. What are the two components of EFE? Give an example of each.
5. What is pragmatic value? When does it dominate action selection?
6. What is epistemic value? When does it dominate action selection?
7. How does Active Inference solve the exploration-exploitation dilemma?
8. Give an example of a situation where you chose to explore (epistemic value) over exploit (pragmatic value).
9. How does motor control work as inference? What is the "prediction" that muscles fulfill?
10. What is proprioception and how does it relate to action as prediction fulfillment?
11. Why are spinal reflexes so fast? How does Active Inference explain this?
12. How does the perception-action loop work? Draw or describe the cycle.
13. How might Parkinson's disease be understood as disrupted action prediction?
14. How does impulsivity relate to insufficient evaluation of Expected Free Energy?
15. How does OCD relate to unresolvable prediction errors about safety?
16. What is an affordance? How does it relate to action selection in Active Inference?
17. Can an agent act without perceiving? Can it perceive without acting? Discuss.
18. How does the concept of a policy (a sequence of actions) differ from choosing one action at a time?
19. In Active Inference, is there a difference between voluntary and involuntary action? Explain.
20. Compare Active Inference's view of action with the traditional stimulus-response model. What are the key differences?